Institutional Configurations and the Social Politics of AI: A Two-Dimensional Typology of OECD Countries

Kaleb Mazurek

April 16, 2026

Abstract

This paper develops a two-dimensional typology of OECD countries that identifies the institutional conditions shaping social exposure to AI. The first dimension captures the type of AI effect a country's workforce faces, displacement or complementarity, measured through task-profile composition from PIAAC. The second captures the depth of labor market dualization, the gap between employment protections for permanent and temporary workers under OECD EPL Version 4. Mapping 29 countries reveals institutional variation that welfare-regime classifications obscure: Spain clusters with the Nordics after its 2012 and 2021 labor market reforms, Sweden's deep formal dualization separates it from the rest of Scandinavia, and the Anglophone countries group with Germany rather than forming a liberal bloc. The theoretical contribution is to recast a contrast the existing literature has not posed. Dualization scholarship emerged around deindustrialization and offshoring, threats to routine manual workers the insider category was built to insulate; AI now disrupts cognitive workers in administrative, legal, and technical roles, the people the insider category was constructed to protect. Whether the same institutional architecture can absorb a threat to its own protected core is the question the typology addresses. Aggregate evidence from the OECD Risks that Matter 2022 survey supports the displacement-complementarity axis directionally; the dualization axis does not predict aggregate preferences, a pattern consistent with the welfare-tradition scope condition identified in the framework. The paper's contribution is the framework itself: two institutional dimensions, a reproducible mapping of OECD countries onto them, and a set of cross-level hypotheses for the individual-level analyses it invites.

Institutional Configurations and the Social Politics of AI: A Two-Dimensional Typology of OECD Countries

Kaleb Mazurek

Independent Researcher, Chicago, IL

Correspondence: kalebmazurek@gmail.com

Working Paper v1.0 | April 2026

JEL Codes: J24, J68, O33, P16

Keywords: artificial intelligence; welfare state; labor market dualization; automation; redistribution preferences; social investment; comparative political economy; OECD

Data and Replication: Country-level data, PCA outputs, cluster assignments, and figure code are available at https://github.com/kmazurek95/typology-paper. The OECD Directorate for Employment, Labour and

Social Affairs provided access to the Risks that Matter 2022 and 2024 Public Use Microdata.

Abstract

This paper develops a two-dimensional typology of OECD countries that identifies the institutional conditions shaping social exposure to AI. The first dimension captures the type of AI effect a country's workforce faces, displacement or complementarity, measured through task-profile composition from PIAAC. The second captures the depth of labor market dualization, the gap between employment protections for permanent and temporary workers under OECD EPL Version 4. Mapping 29 countries reveals institutional variation that welfare-regime classifications obscure: Spain clusters with the Nordics after its 2012 and 2021 labor market reforms, Sweden's deep formal dualization separates it from the rest of Scandinavia, and the Anglophone countries group with Germany rather than forming a liberal bloc. The theoretical contribution is to recast a contrast the existing literature has not posed. Dualization scholarship emerged around deindustrialization and offshoring, threats to routine manual workers the insider category was built to insulate; AI now disrupts cognitive workers in administrative, legal, and technical roles, the people the insider category was constructed to protect. Whether the same institutional architecture can absorb a threat to its own protected core is the question the typology addresses. Aggregate evidence from the OECD Risks that Matter 2022 survey supports the displacement-complementarity axis directionally; the dualization axis does not predict aggregate preferences, a pattern consistent with the welfare-tradition scope condition identified in the framework. The paper's contribution is the framework itself: two institutional dimensions, a reproducible mapping of OECD countries onto them, and a set of cross-level hypotheses for the individual-level analyses it invites.

Keywords: artificial intelligence; welfare state; labor market dualization; automation; redistribution preferences; social investment; comparative political economy; OECD

---

1. Introduction

The past decade has produced increasingly precise measures of AI's effects on labor markets. We can now estimate with reasonable precision which occupations are most exposed and whether exposure leads to displacement or augmentation. Yet we still cannot explain why these measures fail to predict coherent political responses. Workers who perceive higher automation risk support stronger redistribution but express weaker support for social investment (Busemeyer & Sahm, 2023). They want transfers but not retraining.

The divergence is the puzzle. If automation risk simply increased demand for welfare expansion, the implications would be clear: at-risk workers would embrace the forward-looking response that most economists and policymakers advocate. They reject it, and the literature has not given a satisfying account of why. Busemeyer and Tober (2023) tested welfare state generosity as a moderator and found some convergence in the overall level of demand, but a single-dimension treatment cannot explain why the content of that demand varies so sharply across countries at similar generosity levels. Thewissen and Rueda (2019) established that routine task content predicts redistribution preferences; they did not separate types of automation from types

of institutional protection. Findings shift across studies depending on which exposure measure is used, which outcome is examined, and which countries make up the sample.

I argue this inconsistency reflects a missing analytical dimension, not mere measurement noise. Most approaches treat institutional context as a control variable to be partialed out. My argument reverses this: institutions are not background conditions but the primary mechanism through which AI exposure becomes politically consequential. Whether they moderate a stable underlying relationship or constitute it more fundamentally is a question multilevel evidence can adjudicate; the inconsistency of direct effects across the existing literature suggests that a moderation framework alone is incomplete.

The paper's most distinctive theoretical claim concerns the mismatch between what dualization scholarship was built to explain and what AI now does. The insider-outsider architecture emerged to make sense of the institutional fallout from deindustrialization and offshoring, shocks that fell on routine manual workers outside the protected core. AI is a shock to the core itself. Whether the same institutional architecture that once insulated industrial insiders can absorb a threat to cognitive workers in administrative, legal, technical, and middle-management roles, or whether the insider-outsider divide simply migrates upward as the cognitive middle hollows out, is the question the typology is designed to ask.

The measurement literature has made real progress in identifying who is at risk. Felten, Raj, and Seamans (2021) developed the AI Occupational Exposure index. Georgieff and Hyee (2022) adapted it across 23 OECD countries. Pizzinelli et al. (2023) introduced a crucial distinction between substitution and complementarity exposure, which they call complementarity-adjusted exposure. These measures do not specify whether AI will complement or substitute for human labor, which is appropriate at the occupational level. But this leaves political analysis without a country-level construct indicating which effect predominates. The same indices treat workers within an occupation as equally exposed regardless of where they work. A customer service representative in Copenhagen on a permanent contract with retraining rights, and one in Naples with a rolling temporary contract, are not in the same position even if their AIOE scores match. Occupational measures track technological exposure but do not capture social exposure.

A third body of literature bears on the question and has developed largely independently. Emmenegger et al. (2012) showed that labor markets in advanced economies have split into tiers: insiders with strong protections and outsiders with weak or nonexistent coverage. Schwander and Häusermann (2013) took a risk-based approach to outsiderness, defining the divide structurally rather than by current employment, and showed that outsiders demand more redistribution than insiders, even after controlling for income and education. Kurer and Palier (2019) added a temporal dimension: workers anticipating occupational decline shift toward exclusionary politics before displacement arrives.

This paper unites the three literatures through a two-dimensional typology of OECD countries. The first dimension measures the AI effect facing each workforce: displacement or complementarity. This depends on national task composition, which is itself a legacy of prior institutional choices about education systems, training regimes, and economic structure. Nordic economies that invested in human capital now have workforces concentrated in tasks AI augments. Southern European economies with higher shares of routine manufacturing and services face more direct displacement. The second dimension measures how these costs are distributed: the gap between permanent and temporary worker protections. This determines whether

AI's social costs are broadly distributed or concentrated on unprotected outsiders.

No existing paper combines these. Pizzinelli et al. developed the displacement-complementarity distinction but did not connect it to labor market institutions. Häusermann and Schwander developed risk-based outsiderness but applied it to labor-market risk in general, not AI specifically. This paper brings these ideas together and connects them to a political outcome: the type of welfare reform the public demands. That shift makes the question one for political science rather than labor economics.

Crossing the two axes produces four configurations, each with different predicted welfare politics. Where AI is predominantly complementary and protections are broadly shared, the conditions for inclusive social investment coalitions are favorable. Where AI predominantly displaces, and protections concentrate among insiders, outsiders should demand redistribution forcefully but lack the institutional channels to secure it. The other two cells generate patterns that single-dimensional frameworks cannot distinguish.

The paper builds the typology, maps 29 OECD countries onto it using PIAAC task-profile data and OECD employment protection indicators, and tests whether aggregate preference patterns from the OECD Risks that Matter survey are directionally consistent with the predictions.

Section 2 reviews the three literatures and identifies the specific gap the typology fills. Section 3 builds the axes and maps countries onto the space. Section 4 derives testable predictions. Section 5 presents illustrative evidence. Section 6 discusses implications, limits, and the research agenda the framework opens.

---

2. Literature Review

Three literatures bear on this question. They have mostly developed on their own tracks.

2.1 AI Exposure Measurement: What We Can Measure and What We Miss

Work on quantifying AI's labor-market impact has moved quickly. The foundational contribution is the AI Occupational Exposure index by Felten, Raj, and Seamans (2021), which maps progress in specific AI capabilities (image recognition, language modeling, pattern detection) onto occupations based on which human abilities each occupation requires. Occupations that lean heavily on abilities where AI has made the most progress receive higher scores. The index has been widely adopted and extended. Georgieff and Hyee (2022) adapted it for cross-national comparison across 23 OECD countries by mapping AIOE scores onto country-specific occupational structures through ISCO-08 codes. Webb (2020) took a different approach, using patent text to identify which occupations are targeted by AI innovation. Eloundou et al. (2023) assessed exposure to large language models specifically and found that roughly 80 percent of the U.S. workforce has at least 10 percent of their tasks affected by GPT-class models.

These measures share a property worth noting. They are deliberately agnostic about whether AI will complement or substitute for human labor. Felten's AIOE, Webb's patent-based index, and Eloundou's LLM rubric all measure the degree of overlap between AI capabilities and human tasks rather than the direction of the resulting employment effect. That agnosticism is appropriate at the occupational level, because the same capability overlap can produce different outcomes depending on context, and the indices do not pretend

otherwise. But it leaves political analysis without a country-level construct that adjudicates which effect predominates. The political response to augmentation is not the same as the political response to displacement. The first axis of the typology is built to make that adjudication at the country level, using PIAAC task-profile data rather than occupational crosswalks.

A critical advance on the agnosticism problem came from Pizzinelli et al. (2023), who distinguished between substitution and complementarity exposure and developed complementarity-adjusted exposure (C-AIOE). Their framework recognizes that AI exposure is not uniformly threatening. A radiologist whose AI diagnostic tool catches tumors that human eyes miss is highly AI-exposed, but in a way that augments her productivity rather than threatening her employment. A data entry clerk whose AI replacement can process forms faster and without errors is exposed in a way that eliminates the demand for her labor. Pizzinelli et al. found that advanced economies have higher overall exposure than emerging markets, but that the gap narrows substantially when complementarity is accounted for. Within countries, women and highly educated workers face greater exposure of the complementary kind.

These are real advances. They share a limitation that matters for political analysis. All of these indices operate at the occupation level and treat workers within the same occupation as uniformly exposed. Two customer service workers in different countries receive the same AIOE score if their occupational codes match. As a measure of technological exposure, that works. As a measure of social exposure, it does not, because the institutional environment surrounding each worker determines whether technological disruption translates into economic hardship, career redirection, or relative indifference.

The problem runs deeper than missing controls. Lewandowski, Madoń, and Park (2025), using PIAAC data across 108 countries, show that within-occupation task differences explain 78 percent of the cross-country variance in AI exposure; occupational structure accounts for only 22 percent. Within the same occupation, workers in different countries are doing measurably different things. An administrative assistant in Stockholm may spend most of her time coordinating complex stakeholder communications, synthesizing information, and managing cross-functional projects. The same occupational code in Naples may primarily involve data entry, form processing, and routine correspondence. AI augments the first task profile. It can replace the second. PIAAC directly picks up this variation and provides the empirical foundation for the first axis in Section 3. The Lewandowski decomposition holds across a sample well beyond the OECD, strengthening rather than weakening the case for PIAAC-based measurement on the OECD subset.

A further limitation of existing indices: they are silent on who bears the cost. Even if we know a country's workforce is heavily exposed to displacement-type AI, the political consequences depend entirely on how that exposure maps onto the institutional environment. Displaced workers with generous unemployment insurance, retraining programs, and strong dismissal protections elicit a political response sharply different from that of the same workers on temporary contracts with minimal safety nets. The measurement literature tells us who is exposed and how badly. It does not tell us what happens to them when the exposure lands.

2.2 Automation Risk and Welfare Preferences: The Unresolved Divergence

A parallel literature has asked whether automation risk shapes what people want from their welfare states. It does. But the findings contradict each other in ways nobody has satisfactorily explained.

Thewissen and Rueda (2019) established the basic relationship: workers in occupations with higher routine task content express stronger support for redistribution. Their ESS-based analysis showed that occupational risk predicts preferences even after controlling for income, education, and standard demographics. The finding supported the intuition from Meltzer-Richard-type models that economic vulnerability generates demand for compensatory policy.

Busemeyer and Sahm (2023) complicated the picture. They found that individual-level automation risk increases support for redistribution, consistent with Thewissen and Rueda, but simultaneously decreases support for social investment, specifically government spending on education and retraining. The divergence matters politically. Workers most threatened by automation want redistribution, not retraining. They want income protection now, not human capital investment that might pay off later. The finding cuts hard against the social investment paradigm's assumption that workers facing structural change will embrace skills-oriented responses.

Why would at-risk workers reject social investment? The most straightforward explanation is temporal. Retraining and education pay off in three to five years, but workers facing immediate displacement need income now. Credibility matters too. If the labor market offers no good jobs to retrain into, a retraining program looks like paperwork rather than opportunity. Willingness to invest in uncertain future returns also requires a baseline of material security, which is exactly what displacement threatens.

Busemeyer and Tober (2023) added an institutional dimension, testing whether welfare state generosity moderates the micro-level relationship between automation risk and preferences. They found evidence of what the broader literature now calls a welfare buffer effect: in more generous welfare states, the gap between at-risk and non-at-risk workers' preferences narrows, because the baseline of security that generous welfare provides makes automation anxiety less acute for everyone. The welfare buffer effect is real and important. It is also incomplete as an account of cross-national variation.

The incompleteness runs in a specific direction. The welfare buffer effect treats welfare state generosity as a single dimension and asks whether more or less of it matters. It does not address why the content of welfare demand, compensation versus social investment, varies across countries at similar levels of overall generosity. Two countries with comparable social expenditure as a share of GDP can produce very different patterns of public demand if their labor markets are structured differently. A country where AI primarily augments workers should generate different demands from one where AI primarily displaces them, regardless of welfare generosity. A country where the costs of displacement are concentrated on a small outsider population should produce different aggregate preference signals than one where those costs are broadly shared.

The typology below fills exactly this space. Automation risk predicts preferences. Institutional context moderates the relationship. What remains unspecified is which institutional dimensions matter and how they interact. Welfare state generosity is one dimension. The type of AI effect and the distribution of its costs across the insider-outsider divide capture the other two. The divergence between redistribution and social investment that Busemeyer and Sahm documented may not be a universal response to automation risk at all; it may be an artifact of institutional configurations that a single-dimension framework cannot decompose.

The pattern across these studies points toward a stronger claim than any single paper makes. If institutions

were merely moderating a direct link between technological exposure and preference formation, the direct effect should at least be consistently present, varying only in size. It is not. The link appears and disappears depending on measurement choices, country samples, and outcome variables. A moderation framework, in which a stable technology-to-preference relationship is amplified or dampened by context, struggles to accommodate effects that appear and disappear depending on which institutional contexts the analysis includes. Whether institutions instead constitute the relationship more fundamentally is a question multilevel evidence can address; the present analysis cannot.

2.3 Dualization and Political Voice: Who Bears the Cost, Who Can Demand Reform

The third body of literature that matters concerns labor market structure, specifically the divide between insiders and outsiders.

Emmenegger, Häusermann, Palier, and Seeleib-Kaiser (2012) established dualization as a multi-dimensional analytical concept. Modern labor markets in advanced economies have not simply become more unequal over time. They have developed categorical divides between workers with strong institutional protections (permanent contracts, social insurance coverage, collective bargaining representation, access to training) and workers without them (those on temporary contracts, in involuntary part-time work, in platform work, and in other forms of atypical employment). The divide is not just about pay. It determines who can actually access the welfare state.

Schwander and Häusermann (2013) showed that this divide has direct political consequences for redistribution preferences. Their risk-based conceptualization of outsiderness defines the divide structurally, on the basis of biographical and occupational risk of atypical employment or unemployment, rather than on current employment status alone. Defined this way, outsiders express stronger support for compensatory welfare policy than insiders do, even after controlling for standard predictors. This makes sense on its face. Outsiders have less access to social insurance and face more immediate economic vulnerability, so direct redistribution matters more to them than insurance-based programs they cannot access.

Timing matters more than it seems. Kurer and Palier (2019) showed that workers who perceive their occupational status to be declining shift toward exclusionary political attitudes before actual displacement occurs. The political effects of labor market restructuring precede the economic effects. Workers who anticipate automation-driven job loss may change their voting behavior and welfare preferences before they actually lose their jobs. In dualized labor markets, this anticipatory mechanism interacts with the insider-outsider divide: insiders who feel secure may resist reforms that would benefit outsiders at perceived cost to their own protections, while outsiders who feel the threat most acutely lack the organizational channels to turn that anxiety into effective political demand.

Recent qualitative work reinforces the cross-national relevance of these structural differences. Bosmans et al. (2023) interviewed 250 non-standard workers across six European countries. They found that welfare state differences shaped not just employment outcomes but the level of exclusion from social protections, the time scale of insecurity (from daily survival to long-term life planning), and workers' ability to derive any sense of control from their situation. What it feels like to be a labor-market outsider varies enormously across countries in ways that survey data alone does not capture well.

For the politics of AI specifically, the dualization literature identifies a structural problem that neither the measurement nor the preferences literature addresses: the potential mismatch between who needs welfare reform and who can demand it. Outsiders bear a disproportionate share of labor market disruption. They also have lower union membership, lower voter turnout, less organizational representation, and weaker access to institutionalized bargaining. If AI-driven displacement concentrates on the outsider population, the group with the strongest material interest in welfare reform is the group least equipped to achieve it through existing political channels.

One refinement to the insider-outsider framework becomes important when the question shifts to what outsiders actually want from welfare reform, because outsiders are not a monolithic category. High-skilled outsiders, cognitive workers in temporary or project-based employment, tend to favor social investment policy, which Häusermann, Kurer, and Schwander have called ex-ante equalization of opportunities. Low-skilled outsiders favor traditional redistribution. The distinction matters for AI-specific politics because the displacement-complementarity split maps onto this skill gradient: complementarity-exposed economies produce a high-skilled outsider population whose reform preferences lean toward social investment, while displacement-exposed economies produce a low-skilled outsider population whose reform preferences lean toward direct compensation. The typology's predictions in Section 4 rely on this refinement.

The mismatch has implications for the supply side of welfare politics. Palier and Thelen (2010) documented how Continental European unions actively participated in dualization, accepting weaker protections for the periphery in exchange for maintaining strong protections for the core membership. If the same dynamic applies to AI-era labor-market restructuring, the institutions that are supposed to represent workers may end up reinforcing the insider-outsider divide under pressure from automation.

Populist parties have learned to exploit the gap. Chueri (2022) identified an emerging welfare paradigm in which populist radical right parties offer a dualistic vision: generous welfare for those they construct as deserving, typically native-born and long-employed insiders, and exclusion for those they do not, including immigrants, youth in precarious work, and recent labor market entrants. In the context of AI displacement concentrated on outsiders, this paradigm offers insiders a politically attractive alternative to universal reform: protect those who already have protections, and draw the deservingness line so that it excludes the people most exposed to technological displacement.

2.4 The Gap

The gap sits at the intersection. The AI exposure measurement literature identifies who faces disruption and the types of disruption they face. The automation-and-preferences literature establishes that disruption shapes political demand, inconsistently and in ways that depend on unmeasured institutional variation. The dualization literature explains why protections are distributed unevenly and why that distribution has political consequences. None of this literature has been connected to the others in a way that explains the cross-national divergence in welfare preferences.

The gap is sharper than "three literatures have not talked to each other." Dualization scholarship emerged amid the institutional fallout of deindustrialization and offshoring, which primarily threatened routine manual workers. The insider-outsider architecture it described was designed, more or less explicitly, to insulate

industrial insiders from a foreign-competition shock. AI does not threaten the same workers. The exposed population now includes cognitive workers in administrative, legal, technical, and middle-management roles, many of whom sit inside the protected core the EPL regime was built to defend. Whether EPL protections insulate cognitive workers from AI displacement in the same way they protected industrial insiders from offshoring is an open question. It is not one the existing literature has asked. The next section builds a framework that can.

---

3. The Typology

This section builds a two-dimensional framework for classifying OECD countries according to their institutional positioning relative to AI-driven labor-market change. The first axis captures the predominant type of AI effect a country's workforce faces. The second captures how the costs of that effect are distributed across the labor force. Together, they generate a four-cell space with distinct predictions about welfare reform preferences.

3.1 Axis 1: Task-Profile Composition (Displacement versus Complementarity)

Whether AI predominantly displaces or complements labor in a given country depends on what its workers actually do. The relevant unit is the task, not the occupation. Within the same occupation, workers in different countries may be doing substantially different things, and those differences determine whether AI substitutes for their labor or augments it.

The empirical foundation is strong. Acemoglu and Restrepo (2019) formalized the distinction between displacement effects, in which automation replaces tasks previously performed by humans, and reinstatement effects, in which new technology creates new tasks. Pizzinelli et al. (2023) operationalized this specifically for AI by assigning occupations dual scores, one for substitution exposure and one for complementarity exposure, and deriving complementarity-adjusted exposure (C-AIOE) from their combination. Their analysis showed that advanced economies have higher overall AI exposure than emerging markets, but that the balance between substitution and complementarity varies considerably across them. Lewandowski, Madoń, and Park (2025), using PIAAC across 108 countries, demonstrate that within-occupation task differences explain 78 percent of cross-country variance in AI exposure, while occupational structure accounts for only 22 percent. The relevant variation is not in which occupations exist in a country but in what workers within those occupations actually do.

Task composition does not vary randomly. It is the product of decades of institutional investment in education, training, and economic structure. Nordic countries pursued social investment strategies that emphasized human capital formation, producing workforces concentrated in tasks requiring complex judgment, interpersonal coordination, creative problem-solving, and ICT proficiency (Kim & Ahn, 2020; Iversen & Soskice, 2001). These are tasks that current AI systems tend to augment rather than replace. Southern European economies, shaped by different institutional histories and training regimes, retain larger workforce shares in routine manufacturing tasks and routine service tasks, for which AI substitution is more technically feasible and economically attractive.

This is endogenous, of course. Nordic countries face complementary AI exposure because they have invested for decades in human capital. That endogeneity is part of the argument rather than a problem for it: prior welfare state choices shaped how well-positioned countries are for the AI transition, and the typology is designed to make that institutional legacy visible and measurable.

The first axis is operationalized using PIAAC (Program for the International Assessment of Adult Competencies) data, which directly measures task content across roughly 33 countries. PIAAC is preferred over crosswalks from the U.S.-derived O*NET system because it measures what workers actually do in their national contexts, rather than imputing task content from American occupational profiles. The resulting variable is a continuous country-level measure of task complexity, constructed as the first principal component of three PIAAC skill-use indicators: ICT use at work, numeracy use at work, and reading use at work. Higher values indicate workforces concentrated in complex, multi-dimensional tasks, which the literature associates with AI complementarity (Pizzinelli et al., 2023); lower values indicate workforces concentrated in routine tasks more amenable to AI substitution. The measure is a proxy for the displacement-complementarity balance, not a direct implementation of the Pizzinelli scoring methodology, which assigns occupations separate displacement and complementarity scores based on task-AI capability overlap. The proxy assumption is that countries whose workers perform more complex tasks face predominantly complementary AI exposure, while countries whose workers perform more routine tasks face predominantly displacement exposure. The assumption holds at the aggregate cross-national level for current AI capabilities, though LLM-era advances may complicate it for specific high-complexity tasks.

As external validation, the PCA-derived task complexity index correlates at $r = 0.71$ ($p < .001$) with country-level Felten AIOE scores (Felten et al., 2021) aggregated using Eurostat employment shares by ISCO-08 major group ($N = 21$ European countries; Spearman rho = 0.70, $p < .001$).[1] Countries with more complex task profiles have higher aggregate AI occupational exposure, consistent with Pizzinelli et al.'s finding that advanced-economy AI exposure is predominantly complementary. The proxy assumption is empirically supported.

One caveat before Axis 2: complementarity does not necessarily mean benign employment effects. AI can complement workers by making them more productive while simultaneously being labor-saving. A law firm with AI legal research tools might retain five paralegals instead of fifteen, and the remaining five are more productive than the original fifteen. This is complementarity for the survivors and displacement for the ten who were never hired. The typology tracks the type of AI effect and the resulting political dynamics, not the net employment outcome.

[1] Eight OECD countries in the typology mapping (the United States, Canada, Great Britain, Japan, Korea, New Zealand, Israel, and Chile) lack Eurostat occupation data and are not included in the validation; extending it to these countries would require ILO employment-by-occupation statistics and is left for the journal submission. Spain is a notable outlier in the validation, showing high task complexity on the PCA measure alongside relatively low aggregate AI occupational exposure. The interpretation is that Spain's ICT-intensive digital and tourism sectors coexist alongside a large population of lower-AIOE occupations, producing a task-complexity score that exceeds its overall AI exposure. This is a coherent finding rather than a measurement failure: Spain's task profile is genuinely more complex than its occupational composition alone would suggest, which is exactly what PIAAC is designed to detect and what SOC-based exposure measures miss. A leave-one-out sensitivity check confirms the correlation is not driven by any single country; all 21 jackknifed correlations remain positive and statistically significant. Spain's exclusion produces the largest upward shift, raising r to 0.77, consistent with its position as the documented outlier above.

3.2 Axis 2: The Dualization Gap (Who Bears the Cost)

Even within a country, AI exposure is not evenly distributed. The second axis measures how institutional protections concentrate or diffuse the social costs of AI disruption.

The concept of labor market dualization, as developed by Emmenegger et al. (2012), describes the categorical divide between insiders with strong institutional protections and outsiders with weak or nonexistent coverage. The divide structures access not only to employment security but to the welfare state itself. Permanent workers typically enjoy dismissal protections, social insurance contributions, collective bargaining coverage, and employer-provided training. Temporary workers, involuntary part-timers, gig workers, and those in other forms of atypical employment often lack some or all of these protections, even when performing comparable work.

The dualization gap is operationalized as EPRC minus EPT from OECD EPL Version 4, where EPRC captures protection against individual and collective dismissals for regular contracts, and EPT captures regulation of temporary forms of employment. This measures a specific but consequential dimension of institutional inequality. Two countries can have identical average EPL scores but radically different gaps. Denmark and Italy illustrate the point. Denmark's moderate EPL score reflects broadly distributed, moderate protections for the entire workforce, paired with generous unemployment insurance and active labor market policies. Italy's comparable average score masks extremely strong dismissal protections for permanent employees and minimal protections for a large population of temporary and atypical workers. The average is similar. What workers actually experience is not.

When AI arrives in a country with a narrow dualization gap, adjustment costs are absorbed relatively evenly across the workforce because every worker has access to a baseline of retraining and unemployment support. When AI arrives in a country with a deep gap, firms face strong incentives to concentrate adjustment on the flexible periphery. Renewing a temporary contract costs nothing. Firing a permanent employee with strong dismissal protections is expensive and legally complex. The path of least resistance is to automate the tasks performed by temporary workers first, or simply not renew their contracts as AI makes their roles redundant. Balz (2017) found that strong permanent employment protections actually increase perceived job insecurity among temporary workers, because the very rigidity that protects insiders makes outsiders more disposable.

The most challenging critique of this axis is that it merely reflects occupational sorting: temporary workers are in more routine jobs, routine jobs are more AI-exposed, so the dualization gap is really just measuring who has which job. The defense is that the gap measures differential institutional protection rather than task content. Consider two workers doing identical customer service work in the same call center. One holds a permanent contract with severance protections and retraining rights. The other is on a temporary contract that can simply fail to be renewed. The same technology threatens both. The institutional distance between them is what the EPL gap measures.

Supplementary indicators strengthen the measurement. Eurostat temporary employment shares capture how large the outsider population actually is. OECD social protection data on coverage rates by contract type address the multi-dimensionality concern raised by Emmenegger et al., who argued that dualization operates across employment protection, social insurance, and active labor market policy simultaneously. The EPL gap specifically covers the employment protection dimension, and robustness checks using adjusted de facto

enforcement measures (following Jandrić, 2019) address the concern that formal legislation may not reflect actual practice.

3.3 The Two-Dimensional Space and Country Mapping

Crossing the two axes produces a four-cell space. Each cell represents a distinct institutional configuration with different implications for how AI affects labor markets and how publics respond politically. The cells are defined by median splits on both dimensions, producing four clusters. The theoretical predictions below are derived from the framework; Section 5 assesses whether the empirical mapping and observed preference patterns align with them.

Cell 1: Complementary task profile, narrow dualization gap. Countries whose workforces perform complex, multi-dimensional tasks that AI tends to augment, and where employment protections are relatively evenly distributed. The predicted pattern is moderate redistribution demand and strong social investment demand, because the opportunity framing ("AI helps you work better") is credible for a broad share of the workforce and baseline security is broadly guaranteed. The political dynamic should favor inclusive reform coalitions.

Cell 2: Complementary task profile, deep dualization gap. Countries with complex task profiles where a deep gap between permanent and temporary worker protections concentrates adjustment costs on an outsider periphery. The predicted pattern is split: insiders favor social insurance that protects their current earnings; high-skilled outsiders favor social investment because they have the human capital to benefit from retraining and want ex-ante equalization of opportunities; the remaining outsiders favor direct redistribution. The predicted dynamic is reform stasis, because neither insiders (who lack incentive) nor outsiders (who lack voice and whose preferences are internally split) can form an effective reform coalition.

Cell 3: Displacement task profile, narrow dualization gap. Countries whose workforces are concentrated in tasks AI is more likely to replace, but where employment protections are minimal for everyone, rather than concentrated on insiders. The predicted pattern is high redistribution demand broadly shared across the workforce, because the displacement threat is direct and the institutional infrastructure for retraining is thin. Social investment support should be weak. "Retrain for what?" is a credible question when displacement cuts across occupational categories. The political dynamic should produce widespread anxiety with a relatively unified political voice.

Cell 4: Displacement task profile, deep dualization gap. Countries facing displacement-type AI exposure where the costs concentrate on outsiders with weak protections and limited political voice. The predicted pattern is high redistribution demand concentrated among low-skilled outsiders who cannot effectively demand it through institutional channels, while insiders are relatively insulated and feel less urgency. This configuration is most vulnerable to populist exploitation through dualistic welfare paradigms (Chueri, 2022) that offer generous protections for "deserving" insiders and exclusion for outsiders.

Germany occupies a theoretically important position in this framework. Esping-Andersen classifies it as conservative-corporatist alongside France and Austria. The empirical analysis in Section 5 locates it near the center of the two-dimensional space: a moderately complementary task profile (the precision manufacturing core) and a dualization gap just above the median (the Hartz-era expansion of non-standard employ-

ment). This internal heterogeneity, a complementarity-exposed manufacturing sector sitting alongside a displacement-exposed service periphery, is exactly what the regime-type label hides.

---

4. Predictions

The typology generates predictions at three levels. First, the country-level intercept: average preference patterns across the four cells, testable from aggregate cross-national data. Second, the individual-level slope: how the link between AI exposure and preferences varies across institutional contexts, testable only with multilevel data. Third, political implications about whether expressed preferences translate into effective demand, which preference data alone cannot evaluate.

4.1 Baseline Preference Predictions

A preliminary note on what these predictions can and cannot do. They specify the preference patterns the structural logic of each configuration should produce, holding welfare-attitude traditions constant. Welfare traditions vary across the OECD independently of current institutional structures. Post-communist publics express lower redistribution demand than their inequality levels would predict; decades of state socialism disrupted the connection between economic grievance and demand for state-directed compensation (Alesina & Fuchs-Schündeln, 2007). East Asian coordinated economies show a similar disconnect for different reasons. The predictions identify where structural pressure should build. Whether that pressure translates into expressed preferences depends on whether the cultural infrastructure linking structural position to redistributive demand is intact in a given country. This scope condition, identified here and revisited in Section 5.2, is what the aggregate cluster comparison cannot adjudicate and what individual-level estimation with country-level controls is designed to address.

The first set of hypotheses concerns how aggregate preference patterns vary across the four cells. These are predictions about the intercept: the average level and type of welfare reform demand in each configuration, before accounting for individual-level variation.

H1a: Countries with predominantly displacement-type AI exposure exhibit higher aggregate support for redistribution than countries with predominantly complementarity-type exposure.

H1b: Countries with predominantly complementarity-type AI exposure exhibit higher aggregate support for social investment than countries with predominantly displacement-type exposure.

The logic follows from the Busemeyer and Sahm (2023) divergence finding. Displacement poses a direct economic threat, and the intuitive response is compensation. Complementarity creates a productivity opportunity, and the intuitive response is to invest in the capacity to seize it. The typology's contribution is to predict that this divergence is not a universal response to automation risk but varies systematically with the task-profile composition of each country's labor force.

H2a: Among countries with comparable welfare attitude traditions, those with deeper dualization gaps exhibit higher aggregate redistribution demand than those with narrow gaps, holding the type of AI effect constant.

H2b: Among countries with comparable welfare attitude traditions, those with deeper dualization gaps exhibit lower aggregate social investment demand than those with narrow gaps.

The mechanism involves both composition and credibility. Deeper dualization means a larger share of the population occupies the outsider position, which generates redistribution demand (Schwander & Häusermann, 2013). It also means social investment programs are less credible to the outsider population, who have reason to doubt that retraining and education investments will actually reach them. Insiders already receive employer-provided training and have less need for public social investment. The aggregate result is that deep dualization shifts the balance from social investment toward redistribution, at least among outsiders whose skills are not already positioning them for ex-ante equalization-of-opportunities demands rather than compensatory ones.

4.2 The Cross-Level Interaction: When Does AI Exposure Become Politically Consequential?

The hypotheses below require individual-level data on AI exposure, contract status, and policy preferences merged with the country-level typology. They are not evaluable from the aggregate cross-national comparison in Section 5 and are stated here to make the individual-level analysis design transparent.

H3: The combination of displacement-type AI exposure and deep dualization generates structural pressure toward redistribution demand that exceeds what either dimension alone would predict. This interaction is the core of what the typology adds: a routine service worker on a temporary contract in a Cell 4 economy faces high technological exposure and minimal institutional cushioning simultaneously, and the predicted effect on redistribution demand is amplificatory rather than additive.

H4: The average direct effect of individual AI occupational exposure on welfare reform preferences across all countries is weak or null. If institutions are the primary mechanism through which AI exposure becomes politically relevant, the direct technology-to-preference link should be inconsistent across studies and close to zero when averaged across diverse institutional contexts.

H5: The effect of individual AI occupational exposure on redistribution demand is positive in countries with displacement-type task profiles and weaker or absent in countries with complementarity-type task profiles. In a displacement-heavy economy, being in an AI-exposed occupation feels like a direct threat. In a complementarity-heavy economy, the same occupational exposure carries a different subjective meaning because the dominant sectoral experience is augmentation.

H6: In countries with deep dualization gaps, the effect of individual AI exposure on preferences is amplified for outsiders and dampened for insiders. In countries with narrow gaps, this differential attenuation disappears because protections are more evenly distributed. An outsider in a deeply dualized labor market who learns her occupation is highly AI-exposed faces an immediate and concrete threat; the same information, for an insider in the same country, is filtered through a thicker institutional buffer.

4.3 The Political Voice Paradox

The third set of predictions concerns the gap between preference formation and political effectiveness. Preference data alone cannot test these; they are political implications, not statistical hypotheses.

P1: In Cell 4 configurations (displacement plus deep dualization), the population with the strongest welfare reform preferences has the weakest capacity to achieve it through institutional channels.

Outsiders, who bear the concentrated costs of displacement-type AI, are also the group with the lowest union membership, the lowest voter turnout, the least organizational representation, and the weakest access to institutionalized bargaining (Emmenegger et al., 2012). The structural mismatch between demand intensity and political capacity defines Cell 4. These countries do not just face difficult AI transitions. Their institutional architecture routes costs to the people least equipped to respond politically, and in particular to the low-skilled outsider population whose preferences lean heavily toward direct redistribution rather than the social-investment demands articulated more effectively by high-skilled outsiders.

P2: Cell 4 configurations are most susceptible to exploitation by the populist welfare paradigm.

Chueri (2022) documented how populist radical right parties construct a dualistic welfare vision: generous protections for "deserving" insiders and exclusion for those they define as undeserving. In Cell 4 countries, the outsider population bearing displacement costs overlaps substantially with the groups these parties target for exclusion: young workers in precarious employment, recent labor market entrants, and immigrants in atypical employment. The combination of intense reform needs and weak political voice makes these configurations vulnerable to exclusionary populism. The low-skilled outsider population's preference for direct redistribution rather than skills investment makes the populist offer of targeted transfers more attractive than universalist social investment framings.

P3: In Cell 1 configurations (complementarity plus narrow gap), AI transitions are most likely to generate inclusive reform coalitions.

Where protections are broadly shared, and AI predominantly augments the workforce, the political conditions for social investment coalitions are favorable. Encompassing unions can credibly represent both insiders and the relatively small outsider population, and the social investment framing ("AI is an opportunity if we invest in skills") is credible because most workers' direct experience with AI is augmentative. The high-skilled outsider population that does exist in these economies wants ex-ante equalization of opportunities, which aligns with rather than cuts against the insider preference for social investment, making a broad coalition structurally easier to assemble. The risk of exclusionary politics is lowest because the outsider population is smallest and best protected.

P4: Cell 2 configurations (complementarity plus deep dualization) produce reform stasis rather than either inclusive reform or populist capture.

In these countries, insiders in the complementarity-exposed manufacturing core have no strong incentive to push for reform because AI augments their work, and their protections are secure. Outsiders need reform but lack political voice, and their preferences are internally split between high-skilled outsiders who want social investment and remaining outsiders who want direct redistribution. Neither group has sufficient motivation and capacity simultaneously. The predicted outcome is institutional inertia: the welfare state neither expands toward the outsider population nor contracts, because neither a reform coalition nor a retrenchment coalition can form effectively. This prediction differs from Cell 1 (inclusive reform) and Cell 4 (populist exploitation). It is also a prediction that existing typologies, which group Germany with other Continental European states

as a single category, have no way of generating.

---

5. Illustrative Evidence

This section presents descriptive evidence from the empirical mapping. Twenty-nine OECD countries are positioned in the two-dimensional space using PIAAC Cycle 2 task-profile data (2022–2023) and OECD EPL Version 4 indicators. Preference data from the OECD Risks that Matter 2022 survey is used to assess whether aggregate patterns are directionally consistent with the typology's predictions. The evidence here is illustrative, not confirmatory. The country-level mapping below serves as the structural skeleton for the individual-level analysis in preparation.

5.1 Country Positions in the Two-Dimensional Space

Figure 1 plots 29 OECD countries on the two axes. The x-axis captures task-profile composition, measured as the first principal component of country-level means on ICT use at work, numeracy use at work, and reading use at work from PIAAC (73 percent of variance explained; loadings: ICT = 0.52, numeracy = 0.62, reading = 0.59). Higher values indicate workforces concentrated in complex, multi-dimensional tasks that AI tends to augment. The y-axis captures the dualization gap, measured as EPRC minus EPT from OECD EPL Version 4. Higher values indicate deeper institutional divides between insiders and outsiders. Cluster boundaries are drawn at the sample medians on both dimensions.

All four cells are empirically populated, confirming that the two dimensions are analytically independent and do not collapse onto a single diagonal. The clusters contain the following countries:

- **Cell 1** (complementary task profile, narrow dualization gap): Switzerland, Denmark, Spain, Estonia, France, Norway.
- **Cell 2** (complementary task profile, deep dualization gap): Belgium, Canada, Germany, Finland, Great Britain, Ireland, the Netherlands, New Zealand, the United States.
- **Cell 3** (displacement task profile, narrow dualization gap): Austria, Chile, Hungary, Italy, Korea, Lithuania, Portugal, the Slovak Republic.
- **Cell 4** (displacement task profile, deep dualization gap): Czechia, Israel, Japan, Latvia, Poland, Sweden.

Five countries sit near one or both median boundaries: Estonia (essentially at the task-profile median), Poland and Portugal (at the dualization gap median), and Germany and Sweden (close on both). Their cell assignments should be treated as borderline cases sensitive to small measurement changes; robustness checks using alternative cut points are planned as part of the individual-level analysis and will test whether the aggregate patterns depend on these placements.

Germany sits near the center of the two-dimensional space. Its task profile is moderately complementary, reflecting the precision manufacturing core that retained complex production work even as routine assembly moved abroad. Its dualization gap is just above the median, reflecting the Hartz-era expansion of non-standard employment that opened a substantial divide between standard contract employees and the marginal

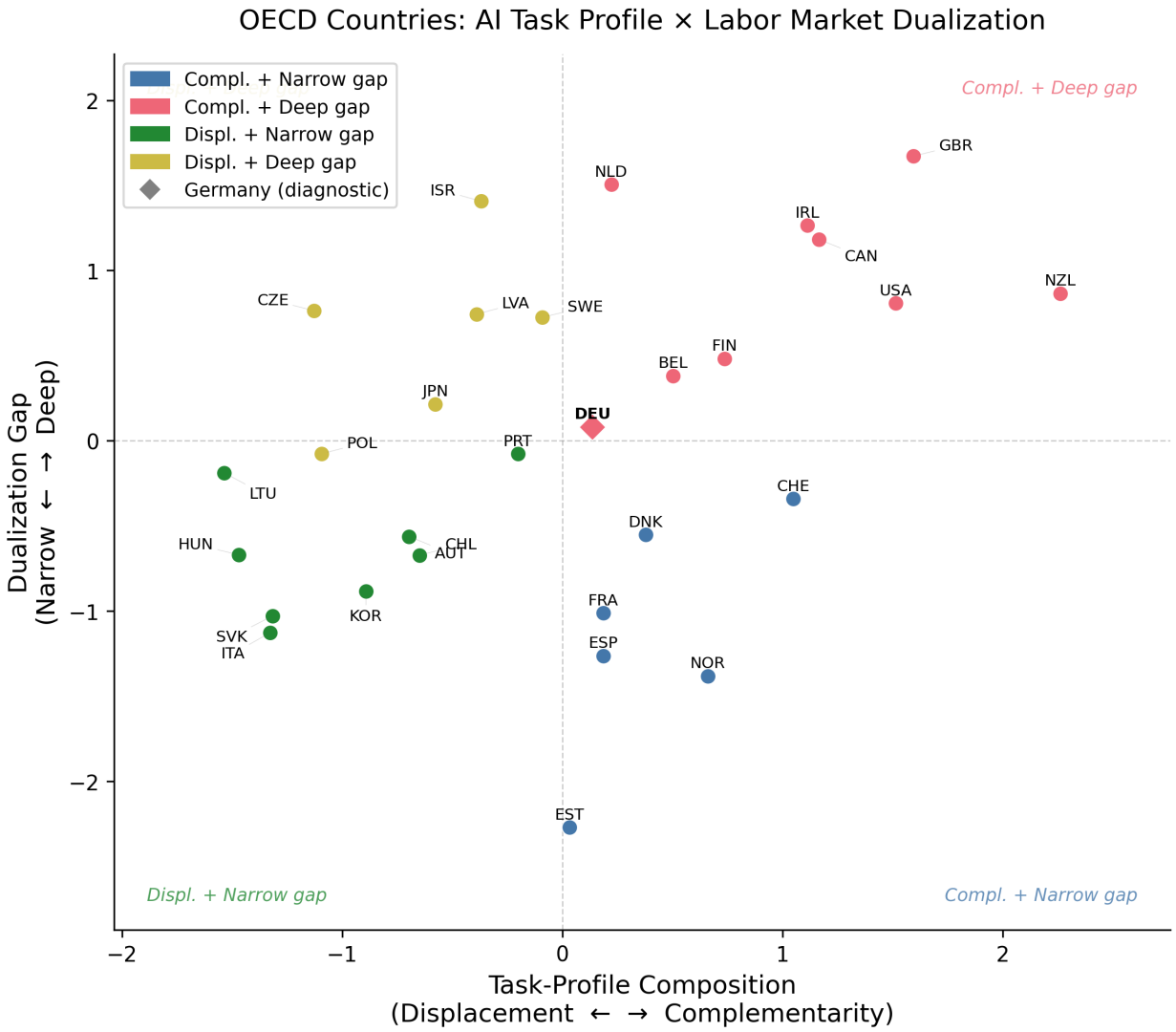

Figure 1: OECD Countries: AI Task Profile × Labor Market Dualization. Cluster boundaries at sample medians. Source: Author's calculations from PIAAC Cycle 2 and OECD EPL Version 4.

employment, mini-job, and temporary agency segment. The two dimensions identify a structural tension the regime label cannot: AI augments work in the manufacturing-export core where insider protections are strongest, while displacing work in the service-sector periphery where outsider exposure has been institutionally manufactured. The reform stasis prediction in Section 4.3 follows from this configuration directly. Insiders in the augmented core have no incentive to push for reform; outsiders in the displaced periphery lack the channels to demand it. The political consequence is not retrenchment and not inclusive expansion, but inertia.

Several other country placements diverge from what existing welfare regime typologies would predict.

Spain's position in Cell 1 is the most counterintuitive. The comparative welfare state literature has traditionally grouped Spain with Italy, Greece, and Portugal. Its placement here reflects two institutional shifts the data picks up directly: the 2012 and especially the 2021 labor market reforms, which converted millions of temporary contracts to permanent ones and compressed the EPL gap substantially, and the structural shift toward ICT-intensive service sectors (digital services, tourism technology, fintech) that pushed the task-profile score upward. Spain in 2023 is genuinely different from Spain in 2008. Static regime classifications miss this. The two-dimensional mapping does not.

Sweden's position in Cell 4 reveals a tension within the flexicurity model that the Nordic label obscures. Sweden's task-profile score falls just below the median, making it a borderline case on Axis 1. But on Axis 2, Sweden has one of the deepest dualization gaps in the sample. Swedish employment protection for permanent workers under the LAS (Employment Protection Act) is among the strictest in the OECD, while temporary workers face considerably weaker protections. The flexicurity narrative emphasizes the generous active labor market policies and unemployment insurance that cushion Swedish outsiders, but those operate through a different institutional channel than the EPL index captures. Sweden's outsiders are better protected than Polish outsiders through ALMP and UI generosity, not through employment protection legislation. A richer Axis 2 that incorporated social insurance coverage by contract type would pick up this distinction.

The Anglophone countries (Great Britain, the United States, Canada, and New Zealand) land in Cell 2 rather than the predicted Cell 3. On Axis 1, this reflects their genuinely complex task profiles. Large professional service, finance, technology, and knowledge-work sectors push the country-level PC1 scores upward despite significant routine service employment. On Axis 2, the EPL gap is wider than expected because these countries score near zero on temporary contract regulation while maintaining nonzero (if weak) permanent contract protections. This placement highlights a measurement limitation. The EPL index was designed for European-style labor markets. In liberal market economies, the relevant form of dualization operates primarily through employer-provided benefits, social insurance eligibility, and the gulf between full-time benefited employment and gig or contract work, dimensions the EPL index was not designed to capture. Whether the Anglophone countries would shift to Cell 3 under an expanded Axis 2 measure is an open empirical question.

This heterogeneity within Cell 2 has implications for the reform stasis prediction in Section 4.3. The Continental European Cell 2 cases (Germany, Belgium, the Netherlands) match the prediction's underlying logic: deep EPL gaps insulate insiders in complementarity-exposed sectors while leaving outsiders without effective political channels. The Anglophone cases share the cell coordinates but arrive there through a different institutional mechanism. Whether they exhibit the same reform stasis dynamic, or whether their benefits-

based dualization produces different political consequences, is a question the current EPL-based measure cannot resolve. A robustness check restricting the Cell 2 reform-stasis prediction to the Continental European cases would clarify whether the prediction's empirical support depends on the heterogeneous cluster composition.

Italy's position in Cell 3 with a near-zero dualization gap (0.035) reflects the 2015 Jobs Act, which introduced the *contratto a tutele crescenti* and substantially weakened dismissal protections for new permanent hires. The protection differential compressed not because temporary workers gained protections but because permanent workers lost some of theirs. Italy moved in the two-dimensional space because of a specific, datable policy reform, exactly the kind of institutional change the typology should be able to detect.

5.2 Aggregate Preference Patterns

Figure 2 compares aggregate redistribution and education spending preferences across the four clusters using data from the OECD Risks that Matter 2022 survey (23 countries with data in both the RTM and the typology mapping).

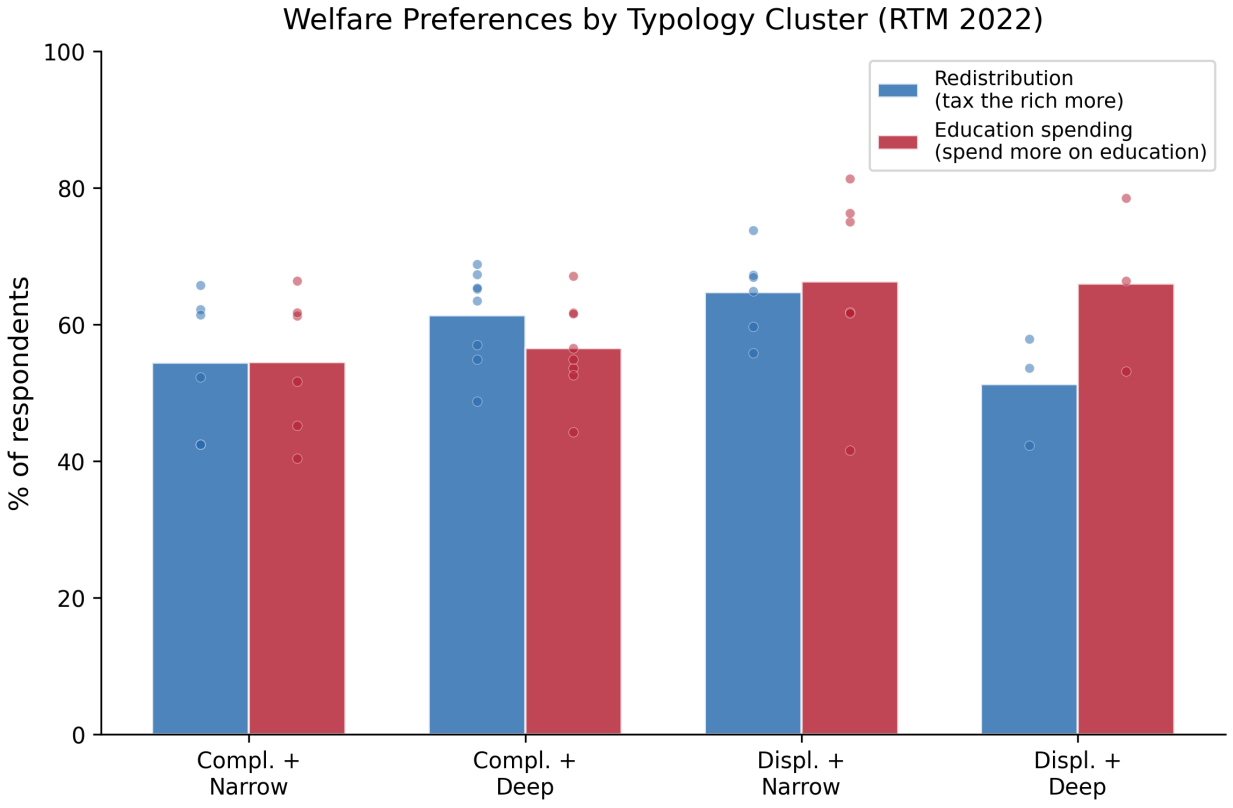

Figure 2: Welfare Preferences by Typology Cluster (RTM 2022). Cluster means with country-level standard errors. Source: Author's calculations from OECD Risks that Matter 2022 Collected Statlinks.

The pattern is partially consistent with the typology's predictions. Cell 3 countries (displacement task profile, narrow gap) show the highest average redistribution demand at approximately 65 percent, consistent with H1a's prediction that displacement-type exposure generates stronger compensatory demand. Cell 1 countries

(complementary task profile, narrow gap) show the lowest average redistribution demand at approximately 54 percent, also consistent with H1a. The ordering across all four cells is Cell 3 > Cell 2 > Cell 1 > Cell 4, which partially supports the displacement-complementarity axis as a predictor of redistribution demand.

The dualization axis does not predict redistribution demand at the aggregate level. Cell 4 (deep gap) shows lower redistribution demand than Cell 3 (narrow gap) among displacement-exposed countries. This is the scope condition made empirical. Cell 4 groups post-communist countries (Czechia, Latvia, Poland), East Asian coordinated economies (Japan, Korea), and one Nordic outlier (Sweden). These countries share structural positions on the two dimensions but differ profoundly in their welfare attitude traditions, and the aggregate comparison cannot separate the structural effect of deep dualization from those baseline tradition differences. Individual-level estimation with country-level controls for welfare tradition is what it takes to adjudicate whether the dualization mechanism works as theorized.[2]

For education spending preferences, the displacement-exposed cells (Cell 3 and Cell 4) show the highest demand, both at approximately 66 percent, while the complementarity-exposed cells (Cell 1 at approximately 54 percent and Cell 2 somewhat higher) show lower demand. This pattern runs opposite to H1b. Two interpretations are consistent with the data. The first is measurement: the aggregate education spending item in the RTM survey may capture general welfare expansion preferences rather than the technology-framed social investment construct H1b actually concerns. The second is substantive: Cell 4 publics may genuinely demand education spending because deep dualization combined with displacement exposure generates demand for any reform that promises mobility, including human capital investment. The substantive interpretation is not implausible, and it would itself be a finding the framework should accommodate. I report the aggregate result here without committing to either reading.

Within-cluster variation is substantial in both figures, as expected given the heterogeneity of countries within each cell. The descriptive cluster comparisons show directional support for the displacement-complementarity axis on redistribution demand. The dualization axis and social investment preferences are less clear at this level of analysis. Whether these patterns sharpen or wash out once individual-level predictors are controlled is the central question for the individual-level multilevel regression to follow.

---

6. Discussion

6.1 What the Typology Reveals

The argument throughout has been that you cannot understand the political consequences of AI-driven labor-market change without specifying the institutional configuration in which that change occurs. The measurement literature, the preferences literature, and the dualization literature each had a piece. Putting them together produces something none of them could generate on their own.

The typology's analytical contribution is to reveal variation that existing welfare regime classifications ob-

[2] One candidate item in the RTM policy battery asks about universal basic income. The broader literature notes that UBI support is concentrated among highly educated, younger respondents rather than among routine workers actually threatened by automation, suggesting the item captures policy attitudes orthogonal to automation-specific self-interest. Subsequent individual-level models will be reported with and without the UBI item as a robustness check.

scure. Esping-Andersen's three worlds framework, for all its enduring influence, groups countries by the generosity and universality of their welfare provisions. For the politics of AI, what matters more are two dimensions Esping-Andersen did not theorize: whether AI creates displacement or complementarity, and whether the costs fall broadly or concentrate on a vulnerable outsider population. The empirical mapping demonstrates this concretely. Spain, traditionally grouped with Southern European welfare states, lands in the same cell as Denmark and Norway because its labor market reforms and service-sector transformation have compressed the dualization gap and shifted the task profile. Sweden, the paradigmatic Nordic case, lands in the same cell as Japan and Poland because its formally deep employment protection dualization is invisible to regime-type labels. The Anglophone countries land with Germany rather than forming a separate liberal cluster, because their task profiles are genuinely complex, even though the form of their dualization (benefits-based rather than EPL-based) is poorly captured by the current measure. None of these placements is an error. They are what the two-dimensional framework is built to detect: institutional variation that regime labels paper over.

The prediction of reform stasis in Cell 2 (complementarity plus deep dualization) deserves particular attention as the framework's most novel implication. Existing frameworks predict either inclusive reform (the social investment literature's optimistic scenario) or welfare retrenchment (the austerity literature's pessimistic scenario). The typology predicts neither. Institutional inertia, because the people who need change cannot demand it, and the people who could demand it have no reason to. If this prediction holds empirically, it has significant implications for understanding why some welfare states fail to adapt to technological change even in the absence of active opposition.

6.2 Limitations

The framework has real limitations, some of which the empirical results make visible.

The country-level sample size is the most consequential methodological constraint. With 29 OECD countries in the current mapping, the cross-level interaction terms specified in H5 and H6 push the limits of what multilevel modeling can reliably estimate. The standard recommendation of 20 to 30 level-2 units applies to random-intercept models. Models with random slopes and cross-level interactions require substantially more. This is a fundamental constraint of comparative welfare state research rather than something specific to this paper, but it means the quantitative analysis must be understood as pattern identification rather than causal estimation. Qualitative case studies of diagnostic cases (a Nordic country, a Southern European country, and Germany as the intermediate case) would provide the mechanism that the quantitative approach alone cannot deliver.

The Employment Protection Legislation indicators capture formal legal provisions, not enforcement practices. Jandrić (2019) showed that country rankings shift meaningfully when de facto enforcement is taken into account. The gap between law and practice may be particularly consequential for outsiders, whose weaker bargaining position makes them less able to enforce whatever formal protections exist. Robustness checks using adjusted measures are essential, but do not fully resolve the concern.

The task-profile data from PIAAC measures task content at a single point in time for most countries. AI capabilities are evolving rapidly, and the boundary between tasks AI can substitute and tasks it can augment

is shifting. Whether the task-profile rankings across countries are stable under changing AI capabilities is an empirical question that repeated PIAAC waves will eventually address. The conceptual framework does not assume task profiles are static, but the specific country positions in the two-dimensional space may shift as both AI capabilities and workforce composition evolve.

The typology is cross-sectional by design. It maps countries at a point in time and generates predictions about preferences at that point. It does not directly model the dynamics of cell-to-cell movement. Countries may shift position as their labor markets restructure, as policy reforms alter the depth of dualization, or as AI technology changes which tasks are exposed. The framework accommodates this conceptually (describing a space through which countries move) but the current empirical approach cannot test temporal dynamics.

The paper treats redistribution and social investment preferences as the primary political outcomes of interest. This is a deliberate scope choice, but it excludes other potentially important responses: support for immigration restriction as a labor market protection strategy, preferences for AI-specific regulation, demand for unconditional basic income, or withdrawal from political participation entirely. The typology could in principle generate predictions about these outcomes. The empirical infrastructure to test them has not been developed here.

The most substantive limitation concerns the welfare tradition confound, which the aggregate results make visible but cannot resolve. Cell 4's preference pattern directly contradicts the structural prediction, driven by the grouping of post-communist and East Asian welfare traditions that have very different baseline attitudes toward state redistribution. Individual-level estimation with country-level controls for post-communist status, welfare expenditure history, and institutional trust levels is what it takes to separate structural from traditional effects. That estimation is the most important extension of the current analysis.

6.3 Research Agenda

Several extensions follow directly from the current analysis.

The most immediate next step is the full multilevel estimation in v2 of this preprint, using pooled OECD Risks that Matter 2022 and 2024 individual-level data merged with PIAAC task profiles and OECD EPL Version 4 indicators. AI exposure at the individual level would be operationalized through NACE industry-based Felten AIOE scores, insider/outsider status through an OECD-derived standard/non-standard worker flag, and the main outcome through a three-item technology-framed policy scale drawn from the RTM digitalization battery. Identification would lean on a wave-by-exposure interaction in the spirit of Bartik-style fixed-treatment difference-in-differences. The specification tests H4–H6 directly by estimating whether individual AI occupational exposure has different preference effects across institutional contexts, controlling for individual-level characteristics and country-level welfare tradition variables. The cross-level interaction approach can separate the structural contribution of the typology's two dimensions from baseline preference differences driven by welfare history, which the aggregate cluster comparison in this paper cannot.

The wave-by-exposure interaction enables a further test the cross-sectional analysis cannot: preference convergence at high exposure levels across institutional contexts. The broader literature suggests that preferences converge across different welfare regimes for individuals facing high-tech-related employment risk, with high-risk workers uniformly prioritizing immediate compensation over social investment. If this conver-

gence holds, the individual-level models should show that the cross-level interactions weaken substantially at the upper tail of the exposure distribution. High-exposure workers in Cell 1 and Cell 4 economies should look more similar to each other than the typology's structural predictions alone would suggest. This is a prediction the typology framework can accommodate without contradicting, because the convergence operates as a ceiling effect on top of the structural pressure the cells identify, rather than as a competing mechanism.

There is also a feedback loop worth investigating. If generous welfare states reduce the supply of cheap, precarious labor (Chueri & Törnberg, 2025), they may paradoxically increase firms' incentives to automate. This mechanism, if operative, means the typology is not static. Cell 1 countries may face accelerating automation pressure precisely because their institutional success removes the cheap-labor buffer that would otherwise absorb it. Testing this requires longitudinal data on AI adoption at the firm or sector level, which is becoming available through OECD and Eurostat enterprise surveys.

The supply side of welfare reform is a separate question entirely. The typology generates predictions about public demand for different reform types, but whether demand translates into actual policy change depends on party system structure, union organizational capacity, government ideology, and the politics of deservingness. Comparative case studies tracing the pathway from preference formation to policy response in diagnostic cases from each cell would complement the quantitative preference analysis.

Union structure may also belong in the framework as a third dimension. Whether encompassing unions can bridge the insider-outsider divide under AI-driven restructuring, or whether they protect their core at the expense of the periphery (as Palier and Thelen documented for earlier reforms), matters for which political trajectory each cell follows. The ICTWSS database provides the institutional data. Measuring the encompassing-versus-fragmented distinction requires qualitative case work beyond what aggregate centralization indices capture.

The rapid evolution of AI capabilities since the release of large language models in late 2022 has changed which occupations feel most exposed. Earlier, AI primarily threatened routine manual and cognitive tasks. LLMs threaten tasks previously considered safely complementary: legal analysis, code writing, content creation, and administrative coordination. Whether this shift alters the task-profile rankings across countries and whether it narrows the complementarity advantage that Cell 1 countries currently enjoy are urgent empirical questions. If LLMs erode the complementarity premium Nordic workforces currently hold, the political dynamics in those countries may shift toward Cell 3 or Cell 4 patterns. The typology would then be most useful not for its static country classifications but for predicting the political consequences of that shift.

---

7. Conclusion

This paper has developed a two-dimensional typology of OECD countries based on the type of AI effect their workforces face and how the costs of that effect are distributed across the labor force. The point of departure was a disconnection in the literature. AI exposure measurement has become increasingly sophisticated, the link between automation risk and welfare preferences remains inconsistent across studies, and the dualization scholarship has not been integrated with either. The typology connects them, and does so by making explicit a contrast the existing literature has not asked: the institutional architecture built to insulate industrial insiders

from offshoring now faces a threat to its own protected core, because AI displaces cognitive workers in administrative, legal, technical, and middle-management roles, the people the original insider category was constructed to protect rather than exclude.

The empirical mapping of 29 OECD countries demonstrates that the framework picks up institutional variation welfare regime classifications miss. Spain lands with the Nordics after its labor market reforms. Sweden's deep formal dualization separates it from the rest of Scandinavia. Italy's post-Jobs Act compression reshapes its position. The Anglophone countries cluster with Germany rather than forming a separate liberal bloc. These placements are not artifacts. They reflect real institutional differences that one-dimensional typologies cannot see. Aggregate preference evidence provides partial support: the displacement-complementarity axis predicts redistribution demand directionally. The dualization axis does not predict preferences at the aggregate level, because welfare traditions vary too much across the OECD for structural position alone to determine expressed attitudes. The scope condition is visible. What makes the framework useful is that it tells us exactly what the individual-level analysis needs to control for and exactly which cross-level interactions to estimate.

The contribution is the framework. Two institutional dimensions, a reproducible method for mapping countries onto them, and a set of testable predictions at the country, individual, and political levels. The places where the descriptive evidence fails are as informative as the places where it holds. The Cell 4 reversal and the aggregate dualization null reveal the boundaries of structural explanation and point directly at welfare tradition as the mediating factor the individual-level analysis needs to address. The typology does not resolve the puzzle it identifies. It clarifies what resolving it requires.

References

Acemoglu, D., & Restrepo, P. (2019). Automation and new tasks: How technology displaces and reinstates labor. *Journal of Economic Perspectives*, 33(2), 3–30. https://doi.org/10.1257/jep.33.2.3

Alesina, A., & Fuchs-Schündeln, N. (2007). Good-bye Lenin (or not?): The effect of communism on people's preferences. *American Economic Review*, 97(4), 1507–1528. https://doi.org/10.1257/aer.97.4.1507

Balz, A. (2017). Cross-national variations in the security gap: Perceived job insecurity among temporary and permanent employees and employment protection legislation. *European Sociological Review*, 33(5), 675–692. https://doi.org/10.1093/esr/jcx067

Bosmans, K., De Roeck, M., Puig-Barrachina, V., Kitiridis, K., De Moortel, D., Greißler, W., Julià, M., Leibold, M., Mauss, D., Paul, K. I., Petri-Romão, P., Ribeiro, A. I., Rompczyk, K. J., Schröder, S., Schubert, T. M., Van Aerden, K., & Vanroelen, C. (2023). Non-standard employment and health and safety in the EU: A systematic review. *International Journal of Environmental Research and Public Health*, 20(5), 4436. https://doi.org/10.3390/ijerph20054436

Busemeyer, M. R., & Sahm, A. (2023). Social investment, redistribution or basic income? Exploring the association between automation risk and welfare state attitudes in Europe. *Journal of Social Policy*, 52(4), 817–840. https://doi.org/10.1017/S0047279421000519

Busemeyer, M. R., & Tober, T. (2023). Dealing with technological change: Social policy preferences and institutional context. *Comparative Political Studies*, 56(7), 968–999. https://doi.org/10.1177/00104140221139381

Chueri, J. (2022). An emerging dualistic welfare paradigm? Unpacking populist radical right parties' welfare state positions. *Scandinavian Political Studies*, 45(4), 402–430. https://doi.org/10.1111/1467-9477.12230

Chueri, J., & Törnberg, P. (2025). How welfare states influence online platform work in Europe. *Journal of European Social Policy*. Advance online publication.

Eloundou, T., Manning, S., Mishkin, P., & Rock, D. (2023). *GPTs are GPTs: An early look at the labor market impact potential of large language models* (arXiv:2303.10130). arXiv. https://arxiv.org/abs/2303.10130

Emmenegger, P., Häusermann, S., Palier, B., & Seeleib-Kaiser, M. (Eds.). (2012). *The age of dualization: The changing face of inequality in deindustrializing societies*. Oxford University Press.

Felten, E., Raj, M., & Seamans, R. (2021). Occupational, industry, and geographic exposure to artificial intelligence: A novel dataset and its potential uses. *Strategic Management Journal*, 42(12), 2195–2217. https://doi.org/10.1002/smj.3286

Georgieff, A., & Hyee, R. (2022). Artificial intelligence and employment: New cross-country evidence. *Frontiers in Artificial Intelligence*, 5, 832736. https://doi.org/10.3389/frai.2022.832736

Iversen, T., & Soskice, D. (2001). An asset theory of social policy preferences. *American Political Science Review*, 95(4), 875–893. https://doi.org/10.1017/S0003055400400079

Jandrić, J. (2019). *De jure and de facto Employment Protection Legislation in post-communist countries: Methodological concerns and measurement* (ILO Working Paper). International Labour Organization.

Kim, S., & Ahn, S. H. (2020). Social investment effects of public education, health care, and welfare service expenditures on economic growth. *Asian Social Work and Policy Review*, 14(1), 34–44. https://doi.org/10.1111/aswp.12193

Kurer, T., & Palier, B. (2019). Shrinking and shouting: The political revolt of the declining middle in times of employment polarization. *Research & Politics*, 6(1). https://doi.org/10.1177/2053168019831164

Lewandowski, P., Madoń, K., & Park, A. (2025). *The role of task content in explaining cross-country differences in AI exposure* (IZA Discussion Paper). Institute of Labor Economics.

Palier, B., & Thelen, K. (2010). Institutionalizing dualism: Complementarities and change in France and Germany. *Politics & Society*, 38(1), 119–148. https://doi.org/10.1177/0032329209357888

Pizzinelli, C., Panton, A., Tavares, M. M., Cazzaniga, M., & Li, L. (2023). *Labor market exposure to AI: Cross-country differences and distributional implications* (IMF Working Paper No. WP/23/216). International Monetary Fund.

Schwander, H., & Häusermann, S. (2013). Who is in and who is out? A risk-based conceptualization of insiders and outsiders. *Journal of European Social Policy*, 23(3), 248–269. https://doi.org/10.1177/0958928713480067

Thewissen, S., & Rueda, D. (2019). Automation and the welfare state: Technological change as a determinant of redistribution preferences. *Comparative Political Studies*, 52(2), 171–208.

https://doi.org/10.1177/0010414017740600

Webb, M. (2020). *The impact of artificial intelligence on the labor market* (Working Paper). Stanford University.

Appendix A: Data and Methods

This appendix documents the data sources, measurement decisions, and processing steps underlying the empirical analysis. The accompanying repository contains the full replication code, raw country-level inputs, and intermediate PCA outputs.

A.1 Data Sources

PIAAC Cycle 2 (2022–2023). Country-level means on three skill-use-at-work scales (ICT, numeracy, reading) were extracted from the OECD Public Use Files (`prg[iso3]p[cycle].csv`). Population-weighted means were computed at the country level using the final plausible-value sampling weights. The Netherlands lacks Cycle 2 data at the time of analysis; Cycle 1 values are used, acknowledged as a cross-cycle imputation in the robustness checks.

OECD Employment Protection Legislation Version 4. Sub-indices for individual and collective dismissals of regular contracts (EPRC, numerically equivalent to EPL_OV) and regulation of temporary forms of employment (EPT) were drawn from the OECD SDMX endpoint for the most recent available year (2019 for most countries). The dualization gap is computed as EPRC minus EPT.

OECD Risks that Matter 2022 Collected Statlinks. Aggregate country-level shares supporting redistribution and education spending were extracted from the public Statlinks tables accompanying the 2022 wave. Twenty-three of the 29 countries in the typology mapping have RTM 2022 coverage.

Eurostat Labour Force Survey (2023 annual). Employment shares by ISCO-08 major group were extracted to construct aggregate Felten AIOE validation values for the 21 European countries with coverage.

A.2 Axis Construction

Axis 1 (task-profile composition). A principal component analysis of the three PIAAC skill-use scales, computed on country-level means, produced a first component accounting for 73 percent of variance with loadings 0.52 (ICT), 0.62 (numeracy), and 0.59 (reading). PC1 scores are used directly as the task-profile score.

Axis 2 (dualization gap). EPRC minus EPT, computed from the OECD EPL Version 4 sub-indices. No transformation beyond the subtraction.

Cluster boundaries. Median splits on both dimensions. Five borderline cases (Estonia, Poland, Portugal, Germany, Sweden) are noted in Section 5.1; robustness tables using $\pm$ 0.1 perturbations of the medians are planned for the individual-level analysis and are not included in this v1 release.

A.3 External Validation of Axis 1

I constructed country-level Felten AIOE scores by taking the employment-weighted mean of Felten ISCO-08 AIOE values across the nine ISCO major groups, using Eurostat Labour Force Survey employment shares. The resulting country-level AIOE was correlated with the PIAAC-derived task-profile score across the 21 European countries with coverage. The Pearson correlation is $r = 0.71$ ($p < .001$); Spearman rho is 0.70 ($p < .001$). The validation uses Eurostat table `lfsa_eisn2`, latest available year, both sexes, age 15+, all NACE sectors summed; `scripts/03b_validate_aioe.py` contains the full specification.

A.4 Aggregate Preference Measurement

The redistribution preference measure is the country-level share of respondents agreeing that the government should do more to reduce income differences (RTM 2022 Statlink item). The education spending preference measure is the country-level share agreeing that the government should spend more on education. These are aggregate descriptive indicators. They are imperfect proxies for the technology-framed policy attitudes the framework ultimately concerns, and the individual-level analysis uses a three-item technology-framed scale drawn from the RTM digitalization battery instead.

A.5 Reproducibility

All data, code, and figure-generation scripts are available at https://github.com/kmazurek95/typology-paper. The repository contains raw country-level inputs (PIAAC means, EPL sub-indices, RTM aggregate shares, Eurostat ISCO shares), the PCA fitting script, the cluster assignment script, and the generation code for Figures 1 and 2.

The repository is dual-licensed: MIT for code, CC-BY-4.0 for the paper, data, and figures. See `LICENSE-CODE` and `LICENSE-DATA-AND-PAPER` in the repository root.

Updated versions of this preprint will be posted to SocArXiv and to the repository's release tags.